NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

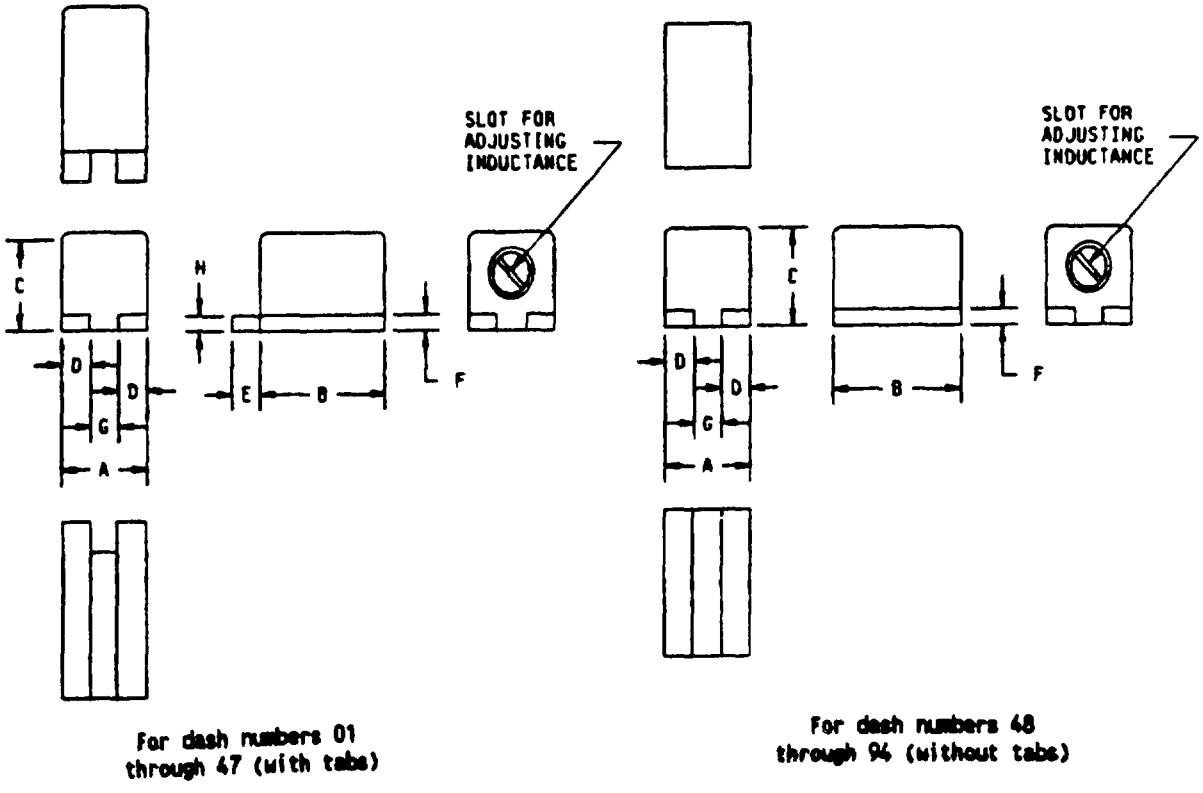
MIL-PRF-83446/7C
6 July 1993
SUPERSEDING
MIL-C-83446/7B
20 February 1985

PERFORMANCE SPECIFICATION SHEET

COILS, RADIO FREQUENCY, CHIP, VARIABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83446.



For dash numbers 01 through 47 (with tabs)

For dash numbers 48 through 94 (without tabs)

FIGURE 1. Variable, chip coil.

(C) denotes changes.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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For dash numbers 01 through 47 (with tabs)		
Ltr	Inches	mm
A	.115 max	2.92 max
B	.170 max	4.32 max
C	.115 max	2.92 max
D	.030 ±.005	0.76 ±0.13
E	.040 +.015 -.010	1.02 +0.38 -0.25
F	.007 ±.003	0.18 ±0.08
G	.038 ±.010	0.97 ±0.25
H	.005 ±.002	0.13 ±0.05

For dash numbers 48 through 94 (without tabs)		
Ltr	Inches	mm
A	.115 max	2.92 max
B	.170 max	4.32 max
C	.115 max	2.92 max
D	.030 ±.005	0.76 ±0.13
F	.007 ±.003	0.18 ±0.08
G	.038 ±.010	0.97 ±0.25

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Variable, chip coil - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 0.5 gram maximum.

Operating temperature range: -55°C to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 700 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202 (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial): See table I.

Electrical characteristics (final): See table II.

Supersession data: A new part numbering system with code for termination materials was incorporated with revision A, superseding MIL-I-83446/7, dated 1 March 1979.

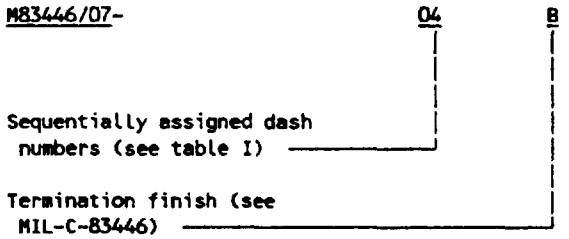
M83446/07-(dash number from table I) supersedes M83446/7-(dash number from table I).

Examples:

M83446/07-21C supersedes M83446/7-21.

M83446/07-03C supersedes M83446/7-3.

(C) Part or Identifying Number (PIN): The PIN shall be in the following form:



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© TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number	Inductance (µH)			Q Min at L nom	Q typ at L nom	Test frequency (MHz) 1/	Self-resonant frequency (min) (MHz)	DC resistance at 25°C (max) (Ohms)	Current (max) (mA) 2/	Incremental current (max) (mA) 3/
	Min	Nom	Max							
01, 48	0.016	0.018	0.019	50	60	150	1200	0.02	750	- - -
02, 49	0.019	0.022	0.024	50	60	150	1100	0.02	750	- - -
03, 50	0.024	0.027	0.029	50	60	150	1000	0.03	750	- - -
04, 51	0.029	0.033	0.036	50	59	150	900	0.03	750	- - -
05, 52	0.036	0.039	0.042	50	60	150	750	0.03	750	- - -
06, 53	0.042	0.047	0.051	50	59	100	660	0.04	750	- - -
07, 54	0.051	0.056	0.062	50	60	100	610	0.04	750	- - -
08, 55	0.061	0.068	0.074	50	58	100	550	0.05	750	- - -
09, 56	0.072	0.082	0.092	50	60	100	520	0.06	640	- - -
10, 57	0.091	0.100	0.114	50	58	50	490	0.07	600	- - -
11, 58	0.112	0.120	0.135	40	50	50	460	0.07	600	- - -
12, 59	0.130	0.15	0.165	40	50	50	430	0.10	480	- - -
13, 60	0.160	0.18	0.195	40	50	50	385	0.12	450	- - -
14, 61	0.195	0.22	0.245	40	51	50	330	0.15	400	- - -
15, 62	0.240	0.27	0.300	40	50	50	275	0.20	350	- - -
16, 63	0.295	0.33	0.365	40	50	50	240	0.25	300	- - -
17, 64	0.360	0.39	0.430	35	46	50	210	0.35	260	- - -
18, 65	0.425	0.47	0.515	35	42	25	190	0.40	250	- - -
19, 66	0.510	0.56	0.620	35	43	25	165	0.50	230	- - -
20, 67	0.610	0.68	0.750	35	42	25	155	0.75	180	- - -
21, 68	0.740	0.82	0.920	35	41	25	145	0.90	160	- - -
22, 69	0.900	1.0	1.10	35	42	25	130	1.8	120	- - -
23, 70	1.10	1.2	1.30	35	43	25	120	1.9	115	- - -
24, 71	1.35	1.5	1.65	22	33	7.9	105	3.0	90	- - -
25, 72	1.60	1.8	1.95	22	32	7.9	95	3.5	85	- - -
26, 73	1.95	2.2	2.45	19	25	7.9	85	6.5	62	- - -
27, 74	2.40	2.7	3.10	19	26	7.9	80	7.0	60	- - -
28, 75	3.00	3.3	3.65	15	24	7.9	75	7.5	55	- - -

See footnotes at end of table.

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© TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number	Inductance (µH)			Q Min at L nom	Q typ at L nom	Test frequency (MHz) 1/	Self-resonant frequency (min) (MHz)	DC resistance at 25°C (max) (Ohms)	Current (max) (mA) 2/	Incremental current (max) (mA) 3/
	Min	Nom	Max							
29, 76	3.60	3.9	4.40	15	22	7.9	70	10.0	50	- - -
30, 77	4.30	4.7	5.20	15	26	7.9	38	2.5	100	- - -
31, 78	5.10	5.6	6.10	20	27	7.9	36	2.5	100	- - -
32, 79	6.10	6.8	7.35	20	28	7.9	35	3.5	85	- - -
33, 80	7.3	8.2	9.4	20	28	7.9	30	4.5	75	- - -
34, 81	9.0	10	11.0	20	28	7.9	25	5.0	70	- - -
35, 82	10.8	12	13.6	20	28	7.9	20	5.5	65	- - -
36, 83	13.4	15	16.8	20	30	2.5	18	6.5	60	- - -
37, 84	16.4	18	20.0	20	30	2.5	17	11.0	47	- - -
38, 85	19.6	22	25.5	20	30	2.5	16	12.0	45	- - -
39, 86	24.5	27	31.0	20	30	2.5	13	13.5	43	- - -
40, 87	30.0	33	38.0	20	30	2.5	11	14.0	42	- - -
41, 88	36.0	39	44.0	20	30	2.5	10	18.0	37	7.5
42, 89	42.0	47	53.0	20	30	2.5	9	20.0	35	7.0
43, 90	50.0	56	64.0	20	30	2.5	8	22.0	34	6.8
44, 91	60.0	68	76.0	20	30	2.5	7	24.0	32	6.4
45, 92	72.0	82	94.0	20	30	2.5	6	38.0	25	5.0
46, 93	90.0	100	115.0	20	28	2.5	5.2	42.0	24	4.8
47, 94	110.0	120	140.0	20	28	2.5	5.0	52.0	22	4.5

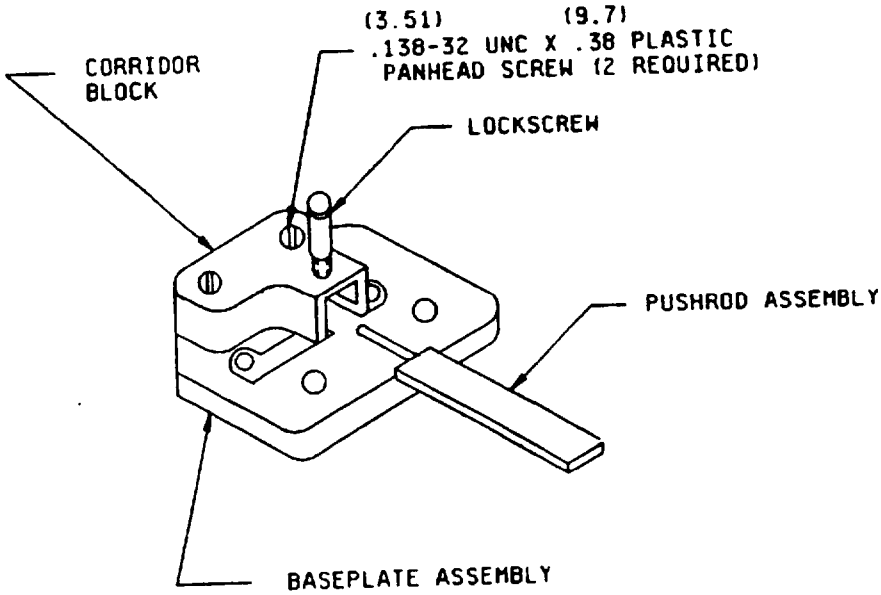
1/ Test frequency range 2.5 through 25 MHz: For electrical characteristics measurements, use the TF-260Q-1 test fixture, or equal (see figure 3). Fixture inductance (approximately 0.028 µH) and residual Q-meter inductance (approximately 0.01 µH) should be subtracted from indicated inductance.

Test frequency range 25 through 150 MHz: For electrical characteristics measurements, use the TF-250RX-1 test fixture, or equal (see figure 2). Fixture inductance (approximately 0.009 µH) should be subtracted from calculated inductance.

2/ Maximum current allowed is not to exceed the specified temperature rise.

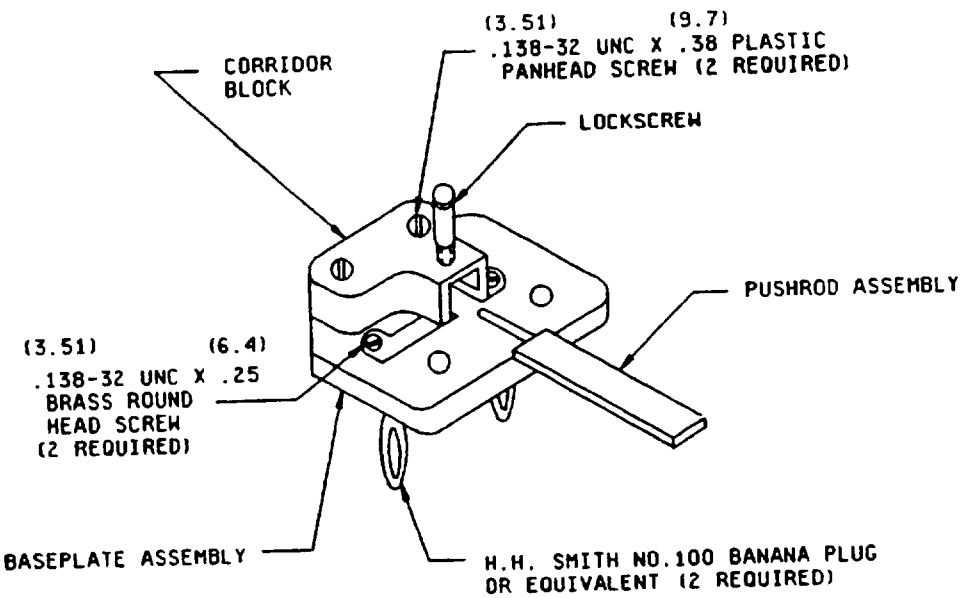
3/ Applicable to ferrite core only.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 2. Typical chip coil test fixture (TF-250RX-1, or equivalent).



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 3. Typical chip coil test fixture (TF-2600-1, or equivalent).

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from the initial measurements			
Qualification inspection	Inductance	DC resistance	Self-resonant frequency	Q
	Percent		Percent	Percent
Group II	±5	±(3% +0.001 ohm)	-8	-10
Group IV	±5	±(2% +0.001 ohm)	-10	-10
Group V	±2	- - - - -	- - -	-10
Quality conformance inspection				
Group C				
Subgroup II	±5	±(3% +0.001 ohm)	-8	-10
Subgroup IV	±5	±(3% +0.001 ohm)	-8	-10

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Navy - AS
Air Force - 19

User activity:
Navy - OS

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0816-02)

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